Core2350B ↔ Tang Nano20K · 16-signal wiring map

Core2350B: RP2350B / 8 MB PSRAM / 16 MB flash • Tang Nano20K hardware 3921

Wiring reservation; physical bus wrapper/driver not implemented
3.3V logic; VGA DAC disconnected

CORE2350B			CABLE / SIGNAL FLOW		TANG NANO20K		
GPIO	Header pin	Signal	At Core	At Tang	Header pin	FPGA pin	Bank
GP10	P1.11	WREADY	33Ω		J6.1	73	1
GP11	P1.14	RVALID	33Ω		J6.2	74	1
GP12	P1.13	ABORT	33Ω		J5.17	72	1
GP13	P1.16	GPU_RESET_N	33Ω		J5.18	71	1
GP14	P1.15	ALE_N	33Ω		J6.15	17	6
GP15	P3.2	RD_N	33Ω		J6.7	16	6
GP16	P3.4	HCLK	33Ω		J6.5	77	1
GP17	P3.3	WR_N	33Ω		J6.6	15	6
GP18	P3.6	D0	33Ω	33Ω	J6.8	27	5
GP19	P3.5	D1	33Ω	33Ω	J6.9	28	5
GP20	P3.8	D2	33Ω	33Ω	J6.12	29	5
GP21	P3.7	D3	33Ω	33Ω	J6.13	30	5
GP22	P3.10	D4	33Ω	33Ω	J6.14	31	5
GP23	P3.9	D5	33Ω	33Ω	J5.5	42	4
GP24	P3.12	D6	33Ω	33Ω	J5.6	41	4
GP25	P3.11	D7	33Ω	33Ω	J5.10	48	3

Power and ground

Power boards through their normal inputs.
Join common GND: Tang J6.20, J5.2, J5.15
to Core P1.1 / P3.1 (fan-out as needed).
Keep inter-board 3.3V and 5V rails separate.
Tang J6.19: local pull-up rail only.
Tang J5.1 (5V), J5.16 (3.3V): unconnected.

Local bias at Tang · 4.7kΩ

To Tang 3.3V: WR_N, RD_N, ALE_N.
To GND: HCLK, ABORT, GPU_RESET_N,
WREADY and RVALID. No data pull resistors.
GPU_RESET_N low holds the GPU in reset.
ABORT is active high; *_N is active low.
Pulls attach on the Tang pin side of series R.

Placement and shared pads

33Ω at each driving end; data: 33Ω each end
(66Ω total). Starting values; timing unqualified.
Disconnect the Tang FPC LCD and VGA DAC.
Preserve HDMI CEC/HPD (FPGA pins 25/26).
3 loads (15/16/17): orange LED + 510Ω to 3.3V.
Future RP drive ≥8mA per net (12mA advised).

Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed.
Assembly while unpowered is OK; keep removable links open on powered old images. 33Ω does not make contention safe.

Core connector numbers are P1/P3 schematic positions, not RP2350 package pins or Pico 2 numbering.

Sources: Core2350B manufacturer schematic; Tang Nano20K Rev 1.30 / hardware 3921 (2023-07-11). Exact evidence: pinout JSON.

Tang Nano20K · physical header map

Hardware 3921 · component side · USB-C TOP / HDMI BOTTOM

Wiring reservation; physical bus wrapper/driver not implemented
3.3V logic; VGA DAC disconnected

J6 = LEFT edge; J5 = RIGHT edge. Both number 1 at top → 20 at bottom.



Blue: Core → Tang • Teal: Tang → Core • Purple: bidirectional data
Grey GPIOs: leave bus wiring off. Orange supply pins: never connect to a Core power rail.
Common ground: J6.20 + J5.2 + J5.15 → Core P1.1 / P3.1. Power each board separately.
FPC LCD disconnected. HDMI CEC/HPD (J6.10/J6.11) remain reserved for HDMI.
Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed.
33Ω does not make contention safe. With old images powered, keep removable links open.

Physical orientation: view the components; do not mirror this map for the solder side. Board outline is schematic, not to scale.

Source: Tang Nano20K Rev 1.30 / hardware 3921 (2023-07-11); full provenance and signal map in the pinout JSON.

Core2350B · P1 / P3 physical locator

RP2350B · 8 MB PSRAM / 16 MB flash · component side · USB FPC connector toward TOP

Wiring reservation; physical bus wrapper/driver not implemented
3.3V logic; VGA DAC disconnected

Use the Core2350B connector labels below; Pico 2 header numbering does not apply.

P1 · LEFT EDGE · vertical 2×8 header

Outer / odd		Inner / even	
P1.1 · GND Common ground to Tang	1	2	**P1.2 · GP0** Existing indicator
P1.3 · GP2 Existing SD SPI	3	4	**P1.4 · GP1** Unallocated by this bus layout
P1.5 · GP4 Existing SD SPI	5	6	**P1.6 · GP3** Existing SD SPI
P1.7 · GP6 Existing audio I2S	7	8	**P1.8 · GP5** Existing SD SPI
P1.9 · GP8 Existing audio I2S	9	10	**P1.10 · GP7** Existing audio I2S
P1.11 · GP10 WREADY · Tang J6.1	11	12	**P1.12 · GP9** Unallocated by this bus layout
P1.13 · GP12 ABORT · Tang J5.17	13	14	**P1.14 · GP11** RVALID · Tang J6.2
P1.15 · GP14 ALE_N · Tang J6.15	15	16	**P1.16 · GP13** GPU_RESET_N · Tang J5.18

P1 contacts read TOP → BOTTOM. Outer column is toward the board's left edge.

P3 · BOTTOM EDGE · horizontal 2×8 header

Read both rows LEFT → RIGHT. Inner / even row is nearer the board centre.

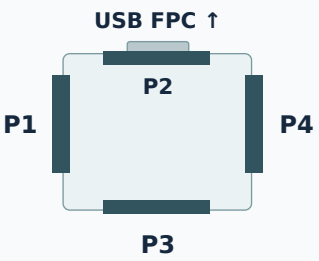
P3.2	P3.4	P3.6	P3.8	P3.10	P3.12	P3.14	P3.16
GP15	GP16	GP18	GP20	GP22	GP24	GP26	GP28
RD_N	HCLK	D0	D2	D4	D6	Unallocated	PS/2 kept
2	4	6	8	10	12	14	16
INNER / EVEN						OUTER / ODD	
1	3	5	7	9	11	13	15
P3.1	**P3.3**	**P3.5**	**P3.7**	**P3.9**	**P3.11**	**P3.13**	**P3.15**
GND	GP17	GP19	GP21	GP23	GP25	GP27	GP29
Common GND	WR_N	D1	D3	D5	D7	PS/2 kept	Unallocated

Physical orientation reference

Component side, USB FPC toward top:
P1 left · P2 top · P3 bottom · P4 right.

Only P1/P3 take bus wiring; grey GPIOs keep existing roles.

Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed; old powered images need links open.
33Ω does not make contention safe. Power boards separately.



Connector locator only. P1/P3 drawings are expanded for legibility and are not a scale board drawing. Do not mirror for solder side.

Source: manufacturer Core2350B schematic and component-side product photograph; full provenance in the pinout JSON.